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ARTICLE

Genetic Population Structure of Bull Trout in the East Fork Bitterroot River Drainage, Montana

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Abstract
Investigation of the genetic population structure of Bull Trout *Salvelinus confluentus* is useful for developing biologically sound conservation and management strategies. We focused on the East Fork Bitterroot River (hereafter, East Fork), Montana, because it is a relatively undisturbed, connected watershed that contains a migratory life history form of Bull Trout. Fin clips were collected from 17 sites: nine East Fork tributaries, the main-stem East Fork, and seven other tributaries across the Bitterroot River drainage. Considering all the population samples, principal components analysis of allele frequencies at 15 microsatellite loci indicated that the East Fork samples formed a cluster that was distinct from the other Bitterroot River tributary samples. Within the East Fork, there was significant divergence among population samples, with pairwise F_{ST} ranging from 0.016 to 0.188 and a global F_{ST} of 0.063. Relative to other Bull Trout studies, levels of genetic variation within our samples were typically higher, while the overall level of differentiation among samples was lower. Based on ONCOR analyses of multiple-locus genotypes, most individuals in the East Fork basin were assigned to their water body of capture, with an average probability of 88%. Within the East Fork, 16 fish that were collected in the main stem were assigned to tributary populations; 26 individuals sampled from seven tributaries were assigned to the main-stem population. In addition, there were five tributaries in which sampled individuals were assigned to tributary populations other than the water body of capture. Based on GeneClass2 analysis, 76 individuals were identified as first-generation migrants. These observations suggest movement and potential gene flow between the main-stem East Fork and its tributaries and between tributaries via the main stem. The main-stem East Fork therefore appears to be an integral component for maintaining the migratory form of Bull Trout in the drainage and may serve as an important vehicle for genetic exchange among tributary populations.

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An understanding of the patterns of gene flow across, within, and among river basins provides information about the degree of isolation among populations of freshwater fishes. This information can be valuable for formulating biologically sound management and conservation programs (Slatkin 1987; Harrison and Hastings 1996; Rieman and Dunham 2000; Lowe and Allendorf 2010). For example, dispersal that leads to gene flow among populations, thus resulting in metapopulation structure, may be particularly important for buffering populations against stochastic environmental risk, supporting sink populations, and refounding extirpated populations (Harrison and Hastings 1996; Hanski 1998; Dunham and Rieman 1999; Rieman and Dunham 2000). In addition, gene flow among spawning populations helps to maintain long-term genetic diversity within populations.

The genetic population structure of the Bull Trout *Salvelinus confluentus* is of particular conservation interest because this species is currently listed as threatened under the Endangered Species Act in the USA (USOFR 1998) and is classified as "at risk" in western Canada (McCart 1997). Most published studies of Bull Trout population structure have found relatively high genetic divergence (F_{ST}; typical range of 0.100–0.250, with differences up to 0.400) not only among populations within large river basins (e.g., Columbia River and Clark Fork River basins) but also among populations in adjacent second- and third-order tributaries (Leary et al. 1993; Kanda and Allendorf 2001; Costello et al. 2003; Spruell et al. 2003; Whiteley et al. 2006; DeHaan et al. 2007, 2010, 2011b; Ardren et al. 2011). Explanations for the high genetic divergence that is commonly observed among proximate Bull Trout populations include their tendency to spawn in headwater streams, their strong site fidelity to natal spawning areas, and their small effective population sizes (N_e), which lead to increased effects of genetic drift (McPhail and Baxter 1996; Swanberg 1997; Neraas and Spruell 2001; Whiteley et al. 2004). Genetic divergence may also be affected by environmental stochasticity, such as debris flows caused by severe wildfires. Such stochasticity may severely reduce the population size and/or isolate the population, thus resulting in the loss of genetic variation, especially rare alleles (Hallerman 2003; Allendorf and Luikart 2007). Anthropogenic activities that modify the landscape (e.g., habitat fragmentation and altered hydrology) can also reduce gene flow and N_e and consequently increase genetic divergence (Neraas and Spruell 2001; Morita and Yamamoto 2002; Yamamoto et al. 2004; DeHaan et al. 2007; Morita et al. 2009).

Much of the existing knowledge about the genetic population structure of Bull Trout has been obtained from studies comparing populations across multiple watersheds or populations within disturbed watersheds, including areas where populations have been fragmented by barriers or habitat degradation (Leary et al. 1993; Kanda and Allendorf 2001; Costello et al. 2003; Spruell et al. 2003; Whiteley et al. 2006; DeHaan et al. 2007, 2008b, 2010). Few published studies have been conducted in relatively undisturbed, connected systems (but see DeHaan et al. 2008a, 2011a, 2011b; Warnock et al. 2010). To ensure a broad understanding of Bull Trout genetic population structure, it is important to compare population structure across an array of systems and scales, as appropriate restoration and conservation goals may differ at these levels.

In addition to an understanding of genetic population structure, the assessment of Bull Trout life history diversity is a conservation priority (Montana Bull Trout Restoration Team 2000). Bull Trout exhibit a variety of life histories, including both resident and migratory individuals that spawn in tributary systems (Rieman and McIntyre 1993; McPhail and Baxter 1996; Northcote 1997). Resident Bull Trout spend their entire lives within their natal stream or tributary, moving only short distances (e.g., <2 km; Jakober et al. 1998). Migratory Bull Trout spend 1–4 years in their natal tributaries before migrating—often tens to hundreds of kilometers—to larger habitats (e.g., rivers, lakes, and estuaries), where they forage and overwinter for several years before returning to the tributaries to spawn (Fraley and Shepard 1989; Rieman and McIntyre 1993; McPhail and Baxter 1996; Swanberg 1997). Migratory Bull Trout are typically over twice the size of resident adults; this larger size greatly increases fecundity and possibly reproductive success (Fraley and Shepard 1989; Jonsson and Jonsson 1993; McPhail and Baxter 1996; Lockard 2006). Populations may be composed of a single life history type, or resident and migratory fish may coexist and breed as a single panmictic population (Jonsson and Jonsson 1993; McPhail and Baxter 1996; Jakober et al. 1998; Nelson et al. 2002; Homel et al. 2008). Conservation of life history diversity, especially migratory forms, is considered important for population persistence in unstable environments. The existence of migratory Bull Trout can lead to gene flow among populations and allow for the possibility of recolonization after catastrophic events (Thorpe 1994; Rieman et al. 1997; Rieman and Dunham 2000; Bahr and Shrimpton 2004; Burton 2005). Despite the importance of maintaining life history diversity (e.g., Schindler et al. 2010), the known threats to Bull Trout population persistence (e.g., habitat degradation, habitat fragmentation, and interactions with exotics) have had disproportionately greater influences on the migratory form than on the resident form throughout the Bull Trout's range (Swanberg 1997; Neraas and Spruell 2001).

Resident and migratory Bull Trout were once common throughout western Montana, and migratory fish were documented regularly in the Bitterroot River drainage throughout the 19th century and early 20th century (M. H. Williams, Bitterroot National Forest, unpublished data). Although Montana Fish, Wildlife, and Parks (MFWP) and the Bitterroot National Forest monitor trout populations in the Bitterroot River drainage, little is known about the genetic population structure of Bull Trout and the status of the migratory life history form (C. G. Clancy, personal communication). Only two areas within the Bitterroot River drainage are still known to support migratory Bull Trout: the West Fork Bitterroot River (hereafter, West Fork) and the East Fork Bitterroot River (hereafter, East Fork). We

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focused on the East Fork because it is a relatively undisturbed, connected, core conservation area for Bull Trout (Montana Bull Trout Scientific Group 1995). Despite these conditions, the Bull Trout migratory life history component has likely been experiencing a decline over the last few decades, as evidenced by decreases in catches of large fish in the main-stem East Fork (Montana Bull Trout Restoration Team 2000; Nelson et al. 2002; MFWP, unpublished data). Our objectives for this study were to (1) describe the genetic population structure of Bull Trout in the East Fork and (2) identify the potential sources of migratory Bull Trout in the drainage. Given results from previous studies, we predicted that neighboring tributaries would be relatively isolated and would have high F_{ST}. We also predicted that fish sampled from the main-stem East Fork would be migratory individuals that would be assigned to several different tributaries. We used an individual assignment test (ONCOR; Kalinowski 2008) and detection of first-generation migrants (GeneClass2; Piry et al. 2004) to help identify which tributaries were producing migratory fish.

STUDY AREA

The Bitterroot River basin is located in southwest Montana and includes the main-stem Bitterroot River, the West Fork, and the East Fork (Figure 1); the total drainage area is 7,288 km^2. The majority of the lower drainage is private land, while higher in the drainage the major landowner is the U.S. Forest Service (USFS). The Bitterroot River basin supports three native salmonids: Westslope Cutthroat Trout *Oncorhynchus clarkii lewisi*, Bull Trout, and Mountain Whitefish *Prosopium williamsoni*. Although the East Fork was our focal area, population samples were included from four tributaries of the main-stem Bitterroot River (Daly, Skalkaho, Burnt Fork, and Willow creeks) and three tributaries of the West Fork (North Fork Sheephead, Sheephead, and Slate creeks) for comparative purposes.

The East Fork headwaters are approximately 73 river kilometers (rkm) from the confluence with the West Fork and main-stem Bitterroot River, with roughly 166 km^2 lying within the Anaconda-Pintler Wilderness (managed jointly by the Beaverhead-Deerlodge and Bitterroot national forests). The watershed encompasses 1,057 km^2. Much of the upper drainage is managed by the USFS, with some private lands being present in the valley bottom. Historical anthropogenic impacts in the watershed include timber harvest, forest roads, agriculture, water diversions, and wildfire. Star Falls (rkm 64), a natural barrier to upstream movement, is the upper limit of known Bull Trout occurrence in the East Fork. There are no known barriers to movement between the tributaries within the watershed below Star Falls, thereby allowing for the expression of both resident and migratory life histories. Biologists from MFWP and USFS have surveyed every tributary of the East Fork, and Bull Trout occupancy has been documented in 17 of the 23 main tributaries. Based on electrofishing data associated with fish monitoring surveys, the number of Bull Trout in the East Fork drainage is estimated to be roughly 5,000–10,000 individuals (MFWP, unpublished data). Tributaries occupied by Bull Trout contain both resident and migratory individuals, but the relative proportions of these life history forms are unknown (MFWP, unpublished data). Other native species that are present in the drainage include the Westslope Cutthroat Trout, Mountain Whitefish, Largescale Sucker *Catostomus macrocheilus*, Longnose Sucker *Catostomus catostomus*, Longnose Dace *Rhinichthys cataractae*, and Slimy Sculpin *Cottus cognatus*. Nonnative fishes that are present in the East Fork include Rainbow Trout *O. mykiss*, Brown Trout *Salmo trutta*, and Brook Trout *Salvelinus fontinalis*. Hybrid fishes found in the East Fork include Westslope Cutthroat Trout × Rainbow Trout, Westslope Cutthroat Trout × Yellowstone Cutthroat Trout *O. clarkii bouvieri* × Rainbow Trout, and Bull Trout × Brook Trout.

METHODS

Sample collection.—During the summers of 2008 and 2009, we electrofished 17 East Fork tributaries where Bull Trout were previously encountered to obtain a population sample (hereafter, "sample" refers to a collection of individuals from a population) from putative spawning populations within the basin (Figure 2). To capture fish, we used either a bank electrofishing system or backpack electrofisher, and we sampled 304.8-m (1,000-ft) sections in an attempt to obtain 50 Bull Trout tissue samples (hereafter, fin clips) representing multiple size-classes. Reaches were added if 50 fin clips were not collected from the initial 304.8-m section. In most tributaries, we were unable to obtain 50 fin clips despite our efforts and the addition of extra sample reaches. We also electrofished approximately 26 km of the main-stem East Fork by using a johnboat mobile anode electrofishing system to capture fish that were residing in the river. The MFWP and USFS provided additional fin clips collected from four tributaries of the main-stem Bitterroot River and three tributaries of the West Fork; these samples allowed us to examine how East Fork genetic diversity fits in a broader geographic scale. Each captured Bull Trout was measured and weighed; a caudal fin clip was collected; and the fish was then released. All fin clips were preserved in vials with a 95% solution of ethanol.

Laboratory methods.—All laboratory work was performed in the Conservation Genetics Laboratory at the University of Montana, Missoula. Analyses of the fin clips included DNA extraction, PCR, and fragment analysis using 15 variable microsatellite loci: *Omm1128* and *Omm1130* (Rexroad et al. 2001); *Sco102*, *Sco105*, *Sco106*, and *Sco107* (Washington Department of Fish and Wildlife, unpublished data); *Sco200*, *Sco202*, *Sco212*, *Sco215*, *Sco216*, *Sco218*, and *Sco220* (DeHaan and Ardren 2005); *Sfo18* (Angers et al. 1995); and *Smm22* (Crane et al. 2004). Seven of the 15 loci are diagnostic between Bull Trout and Brook Trout, thus allowing examination of hybridization between these fishes (Table 1). We extracted DNA from each fin clip by using a cell lysis buffer and ammonium acetate protein precipitation followed by isopropanol DNA precipitation.

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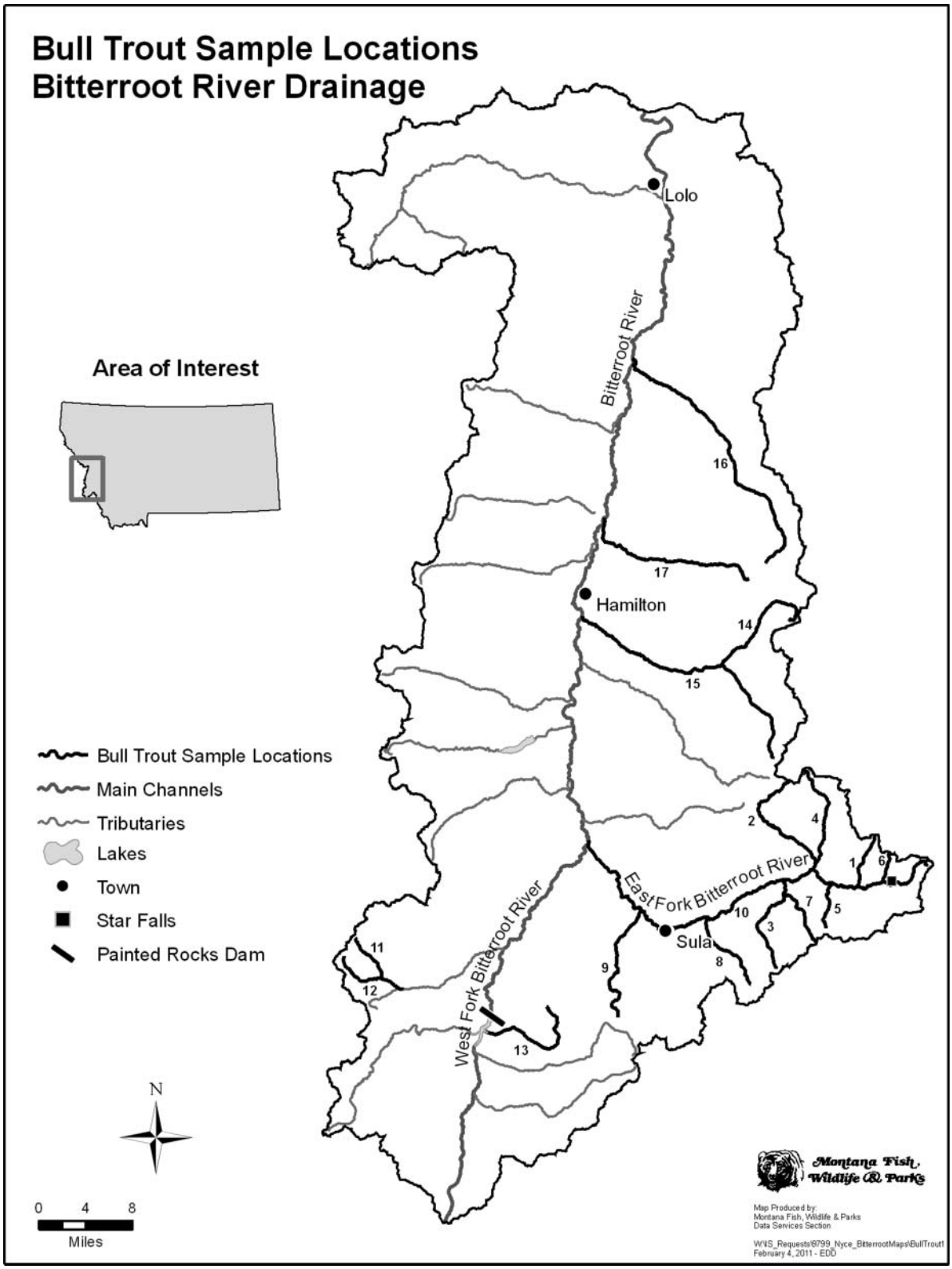

FIGURE 1. Location of Bull Trout sampling sites within the Bitterroot River drainage, Montana: East Fork Bitterroot River (site numbers 1–10), West Fork Bitterroot River (11–13), and Bitterroot River (14–17). Numbers correspond to the numbered populations listed in Table 2.

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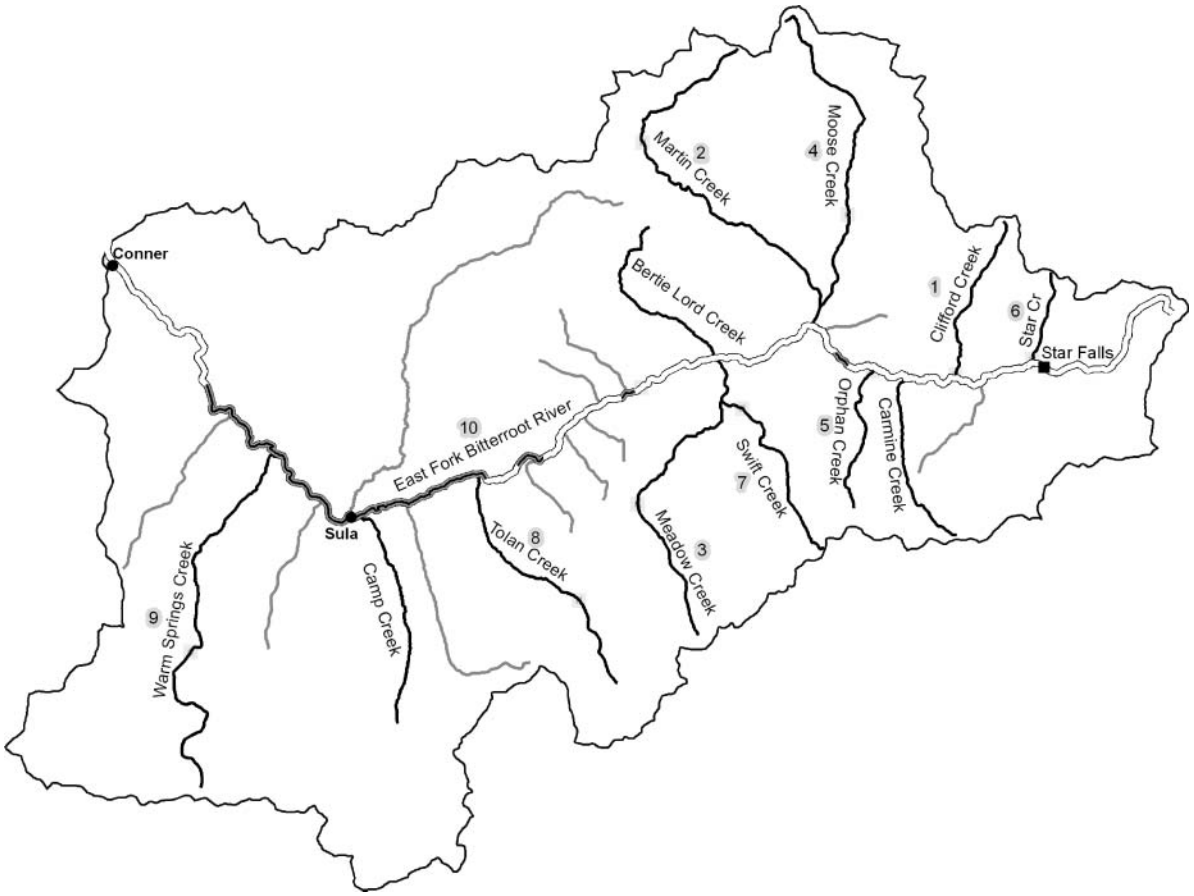

FIGURE 2. Focal study area of the East Fork Bitterroot River drainage, depicting Bull Trout presence and tributaries where tissue samples (fin clips) were collected (highlighted in black). The main-stem East Fork was also sampled to collect fin clips from fluvial fish (highlighted in dark gray). Light-gray highlighting indicates creeks in which no Bull Trout have been captured during any of the Montana Fish, Wildlife, and Parks or U.S. Forest Service sampling efforts to date. Numbers correspond to the numbered populations listed in Table 2.

The PCR reactions were conducted via the QIAGEN microsatellite protocol using the QIAGEN Multiplex PCR Kit (QIAGEN, Valencia, California). We used three different PCR profiles: multiplexes 1 and 2 used a touchdown profile with an initial annealing temperature of 63°C stepping down to 53°C; multiplex 3 used a typical PCR profile with an annealing temperature of 54°C; and multiplex 4 used a typical PCR profile with an annealing temperature of 55.4°C (Table 1). The DNA from each fin clip was amplified in a PTC-200 thermocycler (MJ Research, Waltham, Massachusetts). After PCR, fragments were separated in an ABI 3130xl Genetic Analyzer (Applied Biosystems, Inc. [ABI], Foster City, California) at the Murdock DNA Sequencing Facility, University of Montana, Missoula. Allele sizes were determined by using the ABI GS600LIZ ladder, and chromatogram output was viewed and analyzed using ABI GeneMapper version 3.7.

Genetic analyses.—We performed exact tests of Hardy–Weinberg equilibrium (HWE) for all samples and loci by using the program GENEPOP version 4.0 (Rousset 2008). Significant values for HWE in each sample were adjusted for multiple comparisons by using a sequential Bonferroni adjustment (Rice 1989). We used Fisher's exact test in GENEPOP to test for allele frequency differences between all pairs of samples and thus to determine whether it would be appropriate to consider each sample independently in subsequent analyses. After we established whether samples could be independently considered, we explored the within-sample genetic diversity as well as the magnitude and pattern of among-sample differences.

To examine within-sample genetic diversity, we calculated the mean number of alleles per locus (A), number of private alleles, observed heterozygosity (H_o), and expected heterozygosity (H_e) by using GENALEX version 6.4 (Peakall and Smouse 2006). The program HP Rare version 1.1 (Kalinowski 2005) was used to calculate allelic richness (A_r). This program standardizes the number of alleles detected in each sample to the smallest sample size by using rarefaction. We looked for evidence of recent genetic bottlenecks in the samples by using the program Bottleneck (Cornuet and Luikart 1996), assuming the two-phased model of mutation with a variance of 12.0. Tests for gametic disequilibrium between all pairs of loci in each sample were conducted with Fisher's exact test in GENEPOP.

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TABLE 1. Microsatellite loci analyzed in Bull Trout from the Bitterroot River basin (asterisks indicate loci that are diagnostic for Bull Trout × Brook Trout hybridization); PCR multiplex annealing temperatures (T_A), PCR multiplex final primer concentrations, and associated references (WDFW = Washington Department of Fish and Wildlife).

Locus	Final concentration (μM)	Reference
	Multiplex 1 (T_A = 55°C)	
Sco106	0.10	WDFW, unpublished
*Sfo18**	0.16	Angers et al. 1995
Smm22	0.15	Crane et al. 2004
*Sco216**	0.15	DeHaan and Ardren 2005
	Multiplex 2 (T_A = 56°C)	
*Sco218**	0.10	DeHaan and Ardren 2005
Sco202	0.10	DeHaan and Ardren 2005
Sco200	0.15	DeHaan and Ardren 2005
Sco220	0.12	DeHaan and Ardren 2005
	Multiplex 3 (T_A = 54°C)	
*Sco215**	0.075	DeHaan and Ardren 2005
*Omm1128**	0.10	Rexroad et al. 2001
Sco105	0.10	WDFW, unpublished
	Multiplex 4 (T_A = 55.4°C)	
*Sco102**	0.10	WDFW, unpublished
Omm1130	0.10	Rexroad et al. 2001
*Sco107**	0.10	WDFW, unpublished
Sco212	0.10	DeHaan and Ardren 2005

In addition to allele frequency comparisons, among-sample differences were examined with multivariate analyses and F_{ST}. We used Minitab 15 (Minitab 2007) to perform a principal components analysis (PCA) of the allele frequencies among the samples. Overall genetic divergence among samples (global F_{ST}) was estimated in FSTAT version 2.9.3 (Goudet 2001); the associated 95% confidence level was estimated based on 1,000 bootstrap replicates. To further examine genetic divergence, we computed F_{ST} estimates for all pairs of samples by using GENEPOP following standard ANOVA as described by Weir and Cockerham (1984). Tests of statistical significance were conducted using Arlequin version 3.1 (Excoffier et al. 2005). A sequential Bonferroni correction was used to adjust the significance level for multiple comparisons. To test for a potential isolation-by-distance relationship between F_{ST} and fluvial distance (i.e., rkm between sampling locations), we used Mantel's test in GENALEX (999 permutations).

We used two procedures (individual assignment and detection of migrants) to examine the likelihood of dispersal of individuals among the populations represented by our samples and to identify potential sources of migratory fish. First, we used the leave-one-out and individual assignment options in ONCOR (Kalinowski 2008). The leave-one-out test (a jackknife analysis) was applied to the entire data set to assess the ability of our baseline data set to correctly assign fish to their water body of capture. This procedure involves removal of an individual from the data set and use of a maximum likelihood algorithm to estimate the probability that the individual came from each of the samples; thus, the method was used to ensure that tributary populations in our baseline data set were different enough from each other to permit an informative genetic assignment of fish collected from the main-stem East Fork. After the jackknife analysis, we used the individual genetic assignment test in ONCOR to determine the tributary of origin for each individual fish collected from the main-stem East Fork, as we were interested in identifying sources of migratory fish. Initially, we planned to treat the main-stem East Fork fish as unknowns in the individual assignment analysis because we suspected that they would mainly represent fish that had dispersed from the tributaries. Results of the above analyses, however, suggested that this was unlikely to be the case. Therefore, all 17 samples were included in the individual assignment analysis. Next, we used GeneClass2 (Piry et al. 2004) to search for evidence of potential first-generation migrants among the samples. This analysis used the Bayesian approach of Rannala and Mountain (1997), with 10,000 Monte Carlo-simulated individuals per sample. Designation of an individual as a "migrant" was based on the ratio $L_{\text{home}}/L_{\max}$, where L_{home} is the likelihood computed from the sample in which the individual was collected and $L_{\max}$ is the highest likelihood among all samples (including the sample in which the individual was collected). Individuals were considered to be first-generation migrants (and therefore migratory fish) if the probability of $L_{\text{home}}/L_{\max}$ was less than 0.010. Even though our focus was on the East Fork, we included all 17 samples in this analysis to identify potential migrants from other locations.

RESULTS

Of the 17 East Fork tributaries sampled, Bull Trout were not detected in five tributaries, and three tributaries (Bertie Lord, Carmine, and Camp creeks) were not included in the data set due to very small sample sizes (≤3 fin clips). Thus, samples from nine East Fork tributaries were examined. Each sample included in the data set had at least 15 fin clips (Table 2). Sizes of individual fish in the East Fork tributary samples ranged from 25 to 422 mm TL. In contrast, we captured 73 fish in the main-stem East Fork, with sizes ranging from 102 to 610 mm TL. Individual fish from the three West Fork tributaries and the four Bitterroot River tributaries ranged in size from 76 to 432 mm TL (Table 2). Overall, the final data set included 17 samples across the Bitterroot River basin.

Within-Sample Variation

Our data set was composed of 591 individuals. Among them, 41 possessed both Bull Trout and Brook Trout alleles at the diagnostic loci, indicating that they were hybrids (Table 2). All hybrid individuals occurred in samples from the West Fork and

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TABLE 2. Bull Trout sample information, including the sample size after hybrid fish were excluded (N), the mean number of alleles per locus (A), allelic richness (A_r), expected heterozygosity (H_e), observed heterozygosity (H_o), the number of private alleles, and the number of hybrid individuals.

Sample number	Location	N	A	A_r	H_e	H_o	Private alleles	Hybrids
			East Fork Bitterroot River					
1	Clifford Creek	23	7.133	6.594	0.674	0.704	6	0
2	Martin Creek	35	7.267	5.887	0.624	0.710	0	0
3	Meadow Creek	69	8.267	6.535	0.696	0.718	1	0
4	Moose Creek	38	7.867	6.569	0.660	0.681	0	0
5	Orphan Creek	22	5.267	4.872	0.586	0.685	0	0
6	Star Creek	15	6.133	6.133	0.689	0.756	0	0
7	Swift Creek	50	7.600	5.802	0.628	0.640	0	0
8	Tolan Creek	28	5.467	4.898	0.622	0.650	0	0
9	Warm Springs Creek	27	7.800	6.809	0.680	0.699	1	0
10	East Fork main stem	73	9.467	7.099	0.703	0.700	2	0
Mean		38	7.227	6.120	0.656	0.694	1	0
			West Fork Bitterroot River					
11	North Fork Sheephead Creek	15	6.867	6.867	0.704	0.747	4	4
12	Sheephead Creek	18	7.267	6.922	0.684	0.763	2	1
13	Slate Creek	15	6.933	6.932	0.688	0.756	2	1
Mean		16	7.022	6.907	0.692	0.755	3	2
			Main-Stem Bitterroot River					
14	Daly Creek	51	11.067	8.463	0.734	0.741	3	1
15	Skalkaho Creek	53	10.867	8.142	0.727	0.724	3	6
16	Burnt Fork Creek	32	9.667	8.123	0.751	0.792	10	2
17	Willow Creek	27	6.267	5.352	0.586	0.602	3	26
Mean		41	9.467	7.515	0.700	0.715	5	9

Bitterroot River tributaries, and the majority of these individuals were collected in Willow Creek, a Bitterroot River tributary (Table 2). Hybrids were removed before further analyses, as the presence of nonnative alleles can distort population structure estimates (Forbes and Allendorf 1991; Cegelski et al. 2006; Allendorf and Luikart 2007).

For the 15 microsatellite loci analyzed, A ranged from 2 to 34, and all loci were polymorphic in all 17 samples. After a sequential Bonferroni correction (15 comparisons/sample; $\alpha = 0.003$), there was little evidence of deviations from HWE expectations in the samples. Only Orphan Creek (a tributary of the East Fork) showed a significant departure from HWE, and this was due to an excess of heterozygotes at *Smm22* and *Omm1130*. Although most samples did not show a significant departure from HWE, there was a significant tendency for our East Fork samples to have an excess of heterozygotes (94 of 150 comparisons; Statistical Package for the Social Sciences, sign test: $P = 0.002$). Interestingly, these results include the main-stem East Fork sample, which we initially assumed would consist of a mixture of individuals from different tributaries and thus would possess a deficit of heterozygotes based on HWE. This, however, was not the case. After correction, Fisher's exact test indicated that allele frequencies statistically differed between all pairs of samples at one or more loci. Furthermore, all pairwise F_{ST} estimates were significantly greater than zero ($P < 0.010$; see below and Table 3). Therefore, each sample was treated separately for subsequent analyses.

Comparing all samples in the data set, A ranged from 5.267 to 11.067 with a mean of 7.717 (Table 2). Values of A_r ranged from 4.872 to 8.463, with a mean of 6.587 (Table 2). Values of A and A_r were lowest in Orphan Creek and highest in Daly Creek. Thirty-seven private alleles occurred in 11 different samples, and the frequency ranged from 0.007 to 0.156. Averaged over all samples, H_e and H_o were 0.673 and 0.710, respectively, and the average within-population expected heterozygosity (H_s) ranged from 0.586 to 0.751. Orphan and Willow creeks had the lowest H_e, while Burnt Fork Creek had the highest H_e. Willow Creek had the lowest H_o, and Burnt Fork Creek had the highest H_o. Considering all samples (East Fork, West Fork, and main-stem Bitterroot River), values of A, A_r, and H_e did not significantly differ among populations from the different drainages (Kruskal–Wallis analysis: $P > 0.050$).

Within the East Fork basin, A ranged from 5.267 to 9.467 (mean = 7.227), and A_r ranged from 4.872 to 7.099

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TABLE 3. Estimates of genetic divergence (F_{ST}) between all possible pairs of Bull Trout samples in the data set (East Fork: sample numbers 1–10; West Fork: 11–13; main-stem Bitterroot River: 14–17). All F_{ST} values were significantly greater than zero ($P < 0.01$).

Sample number and location	Sample number															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Clifford Creek																
2. Martin Creek	0.071															
3. Meadow Creek	0.045	0.082														
4. Moose Creek	0.041	0.059	0.032													
5. Orphan Creek	0.103	0.120	0.073	0.073												
6. Star Creek	0.043	0.067	0.047	0.048	0.095											
7. Swift Creek	0.068	0.098	0.043	0.060	0.085	0.065										
8. Tolan Creek	0.124	0.165	0.105	0.116	0.188	0.147	0.148									
9. Warm Springs Creek	0.045	0.090	0.041	0.038	0.106	0.041	0.066	0.113								
10. East Fork main stem	0.016	0.056	0.019	0.018	0.071	0.023	0.030	0.106	0.020							
11. North Fork Sheephead Creek	0.083	0.140	0.076	0.098	0.156	0.089	0.106	0.131	0.074	0.070						
12. Sheephead Creek	0.087	0.122	0.074	0.102	0.161	0.097	0.091	0.126	0.083	0.071	0.041					
13. Slate Creek	0.088	0.114	0.078	0.089	0.139	0.089	0.100	0.122	0.071	0.068	0.046	0.050				
14. Daly Creek	0.080	0.111	0.066	0.072	0.119	0.086	0.092	0.105	0.063	0.060	0.042	0.066	0.050			
15. Skalkaho Creek	0.073	0.109	0.059	0.075	0.119	0.082	0.078	0.093	0.061	0.054	0.042	0.054	0.046	0.009		
16. Burnt Fork Creek	0.097	0.133	0.095	0.096	0.136	0.102	0.130	0.143	0.093	0.081	0.063	0.098	0.079	0.030	0.057	
17. Willow Creek	0.240	0.284	0.212	0.244	0.300	0.227	0.262	0.212	0.239	0.224	0.215	0.208	0.221	0.195	0.179	0.209

(mean = 6.120; Table 2). Orphan Creek had the lowest A and A_r, whereas the East Fork main-stem sample had the highest A and A_r. A total of 10 private alleles occurred in four different samples, and the frequency ranged from 0.007 to 0.130. Averaged over all East Fork samples, H_e and H_o were 0.656 and 0.694, respectively, while H_s ranged from 0.586 to 0.703. Orphan Creek had the lowest H_e, and the main-stem East Fork sample had the highest H_e. Swift Creek had the lowest H_o, and Star Creek had the highest H_o.

We observed evidence of gametic disequilibrium in 8 of the 17 samples. Out of 105 comparisons per sample, the following samples showed significant evidence of disequilibrium: Martin Creek at two pairs of loci, Orphan Creek at three pairs, Star Creek at one pair, Swift Creek at four pairs, Tolan Creek at one pair, North Fork Sheephead Creek at one pair, Sheephead Creek at one pair, and Willow Creek at two pairs. The specific locus pairs that showed evidence of disequilibrium differed among the samples; the exception was observed for *Sco200* and *Sfo18*, which were out of equilibrium in samples from Tolan and Willow creeks. Therefore, disequilibrium was probably not due to linkage but rather to some random factor, such as low N_e or a recent bottleneck. The latter scenario seems unlikely because results of the genetic bottleneck test suggested that only the Star Creek sample had a signal indicating a recent bottleneck, as evidenced by a significant allele frequency mode shift ($P = 0.018$), but the Wilcoxon test was not significant ($P = 0.679$). However, our power to detect a signal indicative of a recent bottleneck was weak due to the relatively small number of individuals and loci analyzed in each sample.

Among-Sample Variation

Comparisons among all samples indicated the presence of geographic genetic structure at the basin level, with the East Fork samples being distinguishable from West Fork and Bitterroot River tributary samples (Figure 3). The first two axes of the PCA explained 43% of the variation in allele frequencies. All

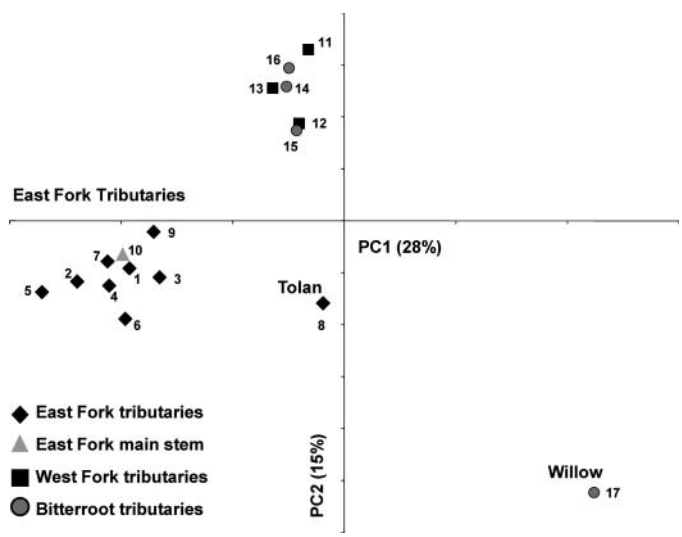

FIGURE 3. Principal components (PC) analysis based on allele frequencies for Bull Trout population samples collected in the Bitterroot River basin. The first two axes (PC1 and PC2) collectively explained 43% of the variation; Tolan and Willow creeks are identified as outliers. Numbers correspond to the numbered populations listed in Table 2.

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but one of the East Fork samples clustered together, and all but one of the samples outside of the East Fork (i.e., the West Fork and Bitterroot River tributaries) clustered together. Both Willow Creek (a Bitterroot River tributary) and Tolan Creek (an East Fork tributary) showed up as outliers from the two clusters (Figure 3). This genetic structure across the basin was also evidenced by pairwise F_{ST} estimates, which ranged from 0.009 to 0.300 (Table 3), with a global F_{ST} value of 0.089. The highest pairwise F_{ST} estimates were observed between Willow Creek and all of the other samples, and pairwise values with Tolan Creek were generally above 0.100. As expected, there was more genetic divergence between tributaries among the three basins (East Fork, West Fork, and Bitterroot River) than between tributaries within the East Fork basin. Specifically, F_{ST} values for paired East Fork tributary samples were generally lower than F_{ST} values from comparisons of East Fork tributary samples with West Fork or Bitterroot River tributary samples. We did not detect a significant isolation-by-distance relationship between genetic distance and fluvial distance for the 17 samples distributed across the Bitterroot River basin ($r^2 = 0.009$, $P = 0.250$).

Focusing on the East Fork, we found that all nine tributary samples and the main-stem East Fork sample significantly differed from each other ($P < 0.010$); pairwise F_{ST} estimates ranged from 0.016 to 0.188 (Table 3), with a global F_{ST} value of 0.063. As previously indicated, the greatest differences tended to be between Tolan Creek and the other East Fork samples, and the smallest differences were between the main-stem East Fork and the tributaries. We did not detect a significant isolation-by-distance relationship between genetic distance and fluvial distance among tributaries of the East Fork ($r^2 = 0.017$; $P = 0.285$).

Results of the leave-one-out test suggested that our baseline had high potential for successful assignment of individual Bull Trout to the correct water body of capture (range = 47–100%; mean = 75%). All individuals that were captured in tributaries of the East Fork were assigned to populations within that basin. Fish from the West Fork and main-stem Bitterroot River tributaries also had high assignment success within each specific basin, with the exception of five individuals. The results from ONCOR (individual assignment test) and GeneClass2 (first-generation migrant detection) analyses suggested that based on our samples, movement of individuals among the East Fork main stem and tributaries may be common (Tables 4, 5). Within the East Fork basin, 76 of 380 individuals were identified as first-generation migrants ($P < 0.010$). Individuals in the main-stem East Fork sample were assigned to or identified as first-generation migrants from all East Fork tributaries except Martin and Tolan creeks. Furthermore, individuals in all East Fork tributary samples except Tolan Creek were assigned to or identified as first-generation migrants from the main-stem East Fork. Finally, a number of individuals sampled in the East Fork

TABLE 4. ONCOR results for individual genetic assignment of Bull Trout samples from the East Fork (sample numbers 1–10), West Fork (11–13), and main-stem Bitterroot River (14–17). Sample location is the water body of capture (where the sample was collected); *N* is the number of fish in the sample; underlined values represent the number of individuals that were assigned to their water body of capture; values that are not underlined represent the number of fish that were assigned to each population other than the location of capture; and "Total" is the total number of fish that were assigned to locations other than the capture location. Empty cells indicate a value of zero.

Sample number and location	*N*	Assigned population number																	Total
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
1. Clifford Creek	23	19			1	1					2								4
2. Martin Creek	35		31		1		1				2								4
3. Meadow Creek	69			59						1	9								10
4. Moose Creek	38				35						3								3
5. Orphan Creek	22					21					1								1
6. Star Creek	15	1					12				2								3
7. Swift Creek	50				1		1	41			7								9
8. Tolan Creek	28								28										0
9. Warm Springs Creek	27									27									0
10. East Fork main stem	73	3		5	1		2	3		2	57								16
11. North Fork Sheephead Creek	15											15							0
12. Sheephead Creek	18												18						0
13. Slate Creek	15													15					0
14. Daly Creek	51														45	6			6
15. Skalkaho Creek	53														4	49			4
16. Burnt Fork Creek	32																32		0
17. Willow Creek	27																	27	0

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TABLE 5. GeneClass2 results for detection of first-generation migrants in Bull Trout samples from the East Fork (sample numbers 1–10), West Fork (11–13), and main-stem Bitterroot River (14–17). Sample location is the water body of capture (where the sample was collected); *N* is the number of fish in the sample; underlined values represent the number of individuals that originated from the water body of capture (i.e., were not considered migrants); values that are not underlined represent the number of migrants, listed under the most likely population of origin; and "Migrants" is the total number of migrants in the sample. All migrant values are significantly greater than zero ($P < 0.01$). Empty cells indicate a value of zero.

Sample number and location	*N*	Assigned population number																	Migrants
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
1. Clifford Creek	23	16			1	1	1				4								7
2. Martin Creek	35		27		2		1				5								8
3. Meadow Creek	69			60	1					1	7								9
4. Moose Creek	38			1	29		1			1	6								9
5. Orphan Creek	22			1		19					2								3
6. Star Creek	15	1					8	1			5								7
7. Swift Creek	50				4		1	36			9								14
8. Tolan Creek	28								28										0
9. Warm Springs Creek	27				4					20	3								7
10. East Fork main stem	73	4		2			2	2		2	61								12
11. North Fork Sheephead Creek	15											11	3		1				4
12. Sheephead Creek	18										1	3	13		1				5
13. Slate Creek	15													12	1	2			3
14. Daly Creek	51														43	8			8
15. Skalkaho Creek	53														12	41			12
16. Burnt Fork Creek	32														2		30		2
17. Willow Creek	27																	27	0

tributaries were assigned to or identified as first-generation migrants from other East Fork tributaries. Thus, there appears to be movement (and potential gene flow) from the main-stem East Fork to most tributaries and vice versa as well as movement between the tributaries (Tables 4, 5).

No individuals from the West Fork or main-stem Bitterroot tributaries were assigned to or identified as potential migrants in the East Fork samples. However, one individual that was captured in Sheephead Creek (a West Fork tributary) was identified as a first-generation migrant (GeneClass2) from the main-stem East Fork. Results from ONCOR indicated that movement between Daly and Skalkaho creeks (main-stem Bitterroot River samples) may be common. GeneClass2 results also indicated that there was movement of fish between West Fork tributaries and between main-stem Bitterroot River tributaries. Likewise, five individuals from tributaries of the main-stem Bitterroot were identified as migrants in the West Fork tributaries (Table 5). A few of these very likely represent spurious results, as they would have required the movement of fish above the impassable Painted Rocks Dam.

DISCUSSION

Our study of the East Fork offers a unique view of Bull Trout genetic population structure. This single watershed is estimated to contain 5,000–10,000 Bull Trout and is a relatively undisturbed, open system. Such conditions have not been common in many previous studies on Bull Trout genetic population structure. We found geographic genetic structure among the East Fork, West Fork, and Bitterroot River samples. The East Fork samples demonstrated high within-population diversity and lower among-population diversity. Movement occurred from the main-stem East Fork to its tributaries, from the tributaries to the main-stem East Fork, and between the tributaries, all of which are indicative of metapopulation structure. We detected no conclusive evidence of fish movement between the East Fork basin and tributaries of the West Fork, or main-stem Bitterroot River. Therefore, the main-stem East Fork is vital for maintaining the migratory form of Bull Trout in this basin.

Although it has been suggested that Bull Trout may function as a metapopulation (Dunham and Rieman 1999; Rieman and Dunham 2000), a few genetic population studies have indicated that there may be little if any metapopulation structure (Spruell et al. 1999; Kanda and Allendorf 2001). Studies have also shown that typically, local populations in close geographic proximity (e.g., adjacent tributaries) are very different genetically (Leary et al. 1993; Spruell et al. 1999; Kanda and Allendorf 2001; Whiteley et al. 2006; DeHaan et al. 2010). Many of these studies compared populations across watersheds or within watersheds characterized by degraded or fragmented habitat. Thus, the results of each study may have been greatly influenced by the condition of the watershed in which it was conducted (Costello

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TABLE 6. Summary of comparable studies of Bull Trout genetic population structure, including watershed size (km^2), global F_{ST}, pairwise F_{ST} range, presence of known barriers, number of populations (N), total number of individuals (N_i), mean number of alleles per locus (A), mean allelic richness (A_r), mean expected heterozygosity (H_e), and mean observed heterozygosity (H_o).

Location	Watershed size (km^2)	Global F_{ST}	Pairwise F_{ST} range	Known barriers	N	N_i	A	A_r	H_e	H_o	Reference
East Fork, Montana	1,057	0.063	0.016–0.188	No	10	380	7.23	6.12	0.66	0.70	This study
West Fork and Bitterroot River, Montana	6,231	0.089	0.009–0.300	Yes	7	211	8.42	7.25	0.70	0.73	This study
Warm Springs Creek, Montana	405	0.206	0.121–0.245	Yes	4	123	3.95	3.82	0.52	0.53	DeHaan et al. 2010
Metolius River, Oregon	1,160	0.054	0.005–0.148	No	7	332	6.28	5.65	0.58	0.59	DeHaan et al. 2008a
Swan River, Montana	1,732	0.037	0.008–0.060	No	7	277	8.41	7.4	0.69	0.69	DeHaan et al. 2011a

et al. 2003; Yamamoto et al. 2004; Meeuwig et al. 2010). The evaluation of population structure in different watersheds across scales and levels of impact is critical as we apply our understanding of Bull Trout population structure to the establishment of restoration goals and conservation guidelines.

Our data indicated that populations in tributaries were genetically different from each other, but the pairwise divergence (estimated by F_{ST}) between samples within the East Fork, West Fork, and main-stem Bitterroot River basins generally was not large. For example, the global F_{ST} among the East Fork samples was 0.063. This is lower than the global F_{ST} observed in many other studies of Bull Trout (typical range of 0.100–0.250, with differences up to 0.400; Whiteley et al. 2006; DeHaan et al. 2010; Ardren et al. 2011) except studies that have been conducted in relatively undisturbed, connected habitats of approximately the same size as the East Fork (see Table 6). The lower F_{ST} values for the East Fork and similar basins suggests that connected, relatively undisturbed watersheds containing a migratory life history component allow a greater opportunity for gene flow among Bull Trout populations. This is supported by our findings from the assignment test and the detection of first-generation migrants; those results indicated that within the East Fork basin, movement from the main-stem East Fork to its tributaries, from the tributaries to the main stem, and among the tributaries was common, likely resulting in gene flow. Overall, these results suggest the existence of metapopulation structure in the East Fork basin (e.g., "metapopulations . . . [are] any assemblage of discrete local populations with migration among them . . . regardless of the rate of population turnover": Hanski and Simberloff 1997). This metapopulation structure may be more typical of smaller, connected watersheds with a diversity of life history types; however, few Bull Trout studies have been conducted in such systems (but see DeHaan et al. 2008a, 2011a; Warnock et al. 2010).

Overall, the genetic diversity we observed within Bull Trout populations in the Bitterroot River drainage tended to be higher than that of Bull Trout in other undisturbed, connected systems of similar size, although there are few studies available for comparison. Comparable studies (Table 6) that used the same 15 microsatellite loci within a single watershed were conducted in Warm Springs Creek, Montana (DeHaan et al. 2010); the Metolius River, Oregon (DeHaan et al. 2008a); and the Swan River, Montana (DeHaan et al. 2011a). Estimates of mean A, A_r, H_e, and H_o were generally higher in the East Fork, West Fork, and Bitterroot River samples than in samples from these other systems, with the exception of the Swan River (Table 6). We can only hypothesize why these measures of genetic diversity were generally greater in the Bitterroot River watershed. If tributary populations only have a few breeders, then an overestimation of population divergence may be caused by sampling the progeny of relatively few breeders (Allendorf and Phelps 1981; Hansen et al. 1997; Hansen and Mensberg 1998). Sampling methods, population age structure, and breeding ecology could vary among studies, thereby influencing the estimates of divergence. A more likely explanation is that the Bitterroot River watershed may have higher population sizes and/or greater connectivity (lack of barriers and number of first-generation migrants) than the other systems. Higher proportions of migratory fish and increased habitat size and quality in the East Fork may also contribute to the observed differences (Hansen and Mensberg 1998; Knutsen et al. 2001; Neville et al. 2006). We suggest that the relatively high within-population genetic diversity is also a reflection of the higher levels of gene flow among populations in such undisturbed, connected systems. This idea is supported by studies that have demonstrated a reduction in genetic variation within Bull Trout populations that are isolated above barriers (e.g., Whiteley et al. 2006; DeHaan et al. 2007, 2011b). The authors of these prior studies suggested that the size of Bull Trout populations and the distance between populations with migratory fish could be a factor influencing the levels of genetic variation. Similarly, in Brown Trout, resident populations show greater divergence and lower diversity than populations

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with migratory individuals, thus highlighting the role of life history variation in patterns of genetic variation (Hansen and Mensberg 1998; Knutsen et al. 2001).

In comparing the allele frequencies among all samples from the East Fork, West Fork, and Bitterroot River, we found two major groups of samples: (1) the East Fork, with Tolan Creek as an outlier; and (2) the West Fork and the Bitterroot River, with Willow Creek as an outlier (Figure 3). Even though there are no known barriers between Tolan and Willow creeks and their respective main-stem rivers, these populations appeared to be more isolated than the other populations in the region. It is possible that Tolan and Willow creeks are more isolated because of watershed disturbances, including (but not limited to) hydrology and land use. For example, the lower end (3.2 km) of Tolan Creek is heavily impacted by agricultural operations (grazing and water diversion). The Willow Creek watershed is affected by roads running along the creek, channelization and private development, and grazing and diversion structures that likely reduce dispersal (R. Brassfield and M. Jakober, USFS, personal communication). In addition, Willow Creek had the largest proportion of Bull Trout × Brook Trout hybrids (about 50% of samples), suggesting that there is an abundance of Brook Trout, more suitable habitat for Brook Trout, or fewer Bull Trout. Regardless of the reason, isolation of the two creeks appears to have resulted in lower within-population genetic diversity and unusually high divergence for these populations.

Populations with less genetic diversity, such as those in Tolan and Willow creeks, should not necessarily be treated as having a lower conservation value than the more diverse populations. These populations constitute part of the overall genetic diversity of the system (Rieman and Allendorf 2001). It is also possible that Bull Trout in these two tributaries contain important adaptive differences resulting from the lack of gene flow (Leary et al. 1993; Giles and Goudet 1997). Due to their genetic differences and despite their isolation, Bull Trout in Tolan and Willow creeks may represent important components of the among-population genetic diversity of Bull Trout in the Bitterroot River basin.

In 8 of the 17 samples we examined, there was significant gametic disequilibrium between one or more pairs of loci. Immigrants that differ genetically from local individuals can create gametic disequilibrium, especially if migration rates are low and pulsed or if populations are small (Waples and England 2011). Waples and England (2011) also noted that the same effect could occur if individuals from more than one population are included in the sample. The gametic disequilibrium and the observed tendency toward an excess of heterozygotes could indicate that N_e is relatively low. In contrast, the relatively low estimates of F_{ST}, especially in the East Fork, and the results from the assignment test and migrant detection suggest that there is gene flow among populations. Gene flow among populations with a breeding ecology that promotes a relatively low N_e may be very important for maintaining genetic diversity (Rieman and Allendorf 2001).

The presence of migratory fish and dispersal among populations may be important for the persistence of Bull Trout in the East Fork and throughout the Bitterroot River system. Distribution of adults into multiple habitats (rivers and tributaries) can buffer populations from disturbances. For example, Rieman et al. (1997) and Sestrich et al. (2011) determined that Bull Trout populations recovered rapidly after extensive wildfires. Rieman et al. (1997) concluded that two important mechanisms contributing to Bull Trout recovery were dispersal and life history (migratory fish). Even though wildfire burned extensively throughout the East Fork basin in 2000, with several creeks experiencing high-severity riparian burns and subsequent debris flows (Sestrich et al. 2011; USFS, unpublished data), only the Star Creek sample showed any signs of a recent population bottleneck. However, this drainage experienced very little wildfire, and where it did experience wildfire the severity of the wildfire was low (USFS, unpublished data).

One of our goals was to identify the potential sources of migratory Bull Trout. Even though some fish sampled in the main-stem East Fork were assigned to the tributaries, 78% of fish in the main-stem sample were assigned to that same population and grouped together in HWE. Historical redd count data and radiotelemetry studies have identified some spawning in the upper main-stem East Fork, thus indicating that this area is a potential source of migratory fish (Nyce 2011; MFWP, unpublished data). In the tributary samples, many of the identified first-generation migrants were assigned to the main-stem East Fork. Likewise, in the main-stem East Fork sample, many of the identified first-generation migrants were assigned to the tributaries. These results imply that the main-stem spawning population produces and receives a large proportion of migrants and that the East Fork main stem currently plays an important role for migratory fish. In addition, many of the East Fork tributaries, especially Clifford, Meadow, Star, Swift, and Warm Springs creeks, were identified as sources of migratory fish (Tables 4, 5). Given that we detected first-generation migrants collected over a short time period (2 years), additional sampling across a longer time period could eventually identify other tributaries that are important and that may contribute to the persistence of the migratory life history form.

Although continued protection of Bull Trout populations in the tributaries is essential, the main-stem East Fork appears to be an integral component for maintaining the migratory form of Bull Trout in the drainage. The main-stem East Fork not only appears to produce and receive migratory fish but also serves as a dispersal corridor for potential genetic exchange among tributary populations. Conservation of migratory Bull Trout in the East Fork is critical (USFWS 2008), and although the management unit is the entire East Fork basin, a concerted monitoring effort should be focused on the main-stem East Fork. To help ensure Bull Trout persistence, we would prioritize both the identification of spawning locations in the main-stem East Fork and the preservation of habitat as necessary for migratory Bull Trout.

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